

# COS10005

## Web Development

### Lecture 1

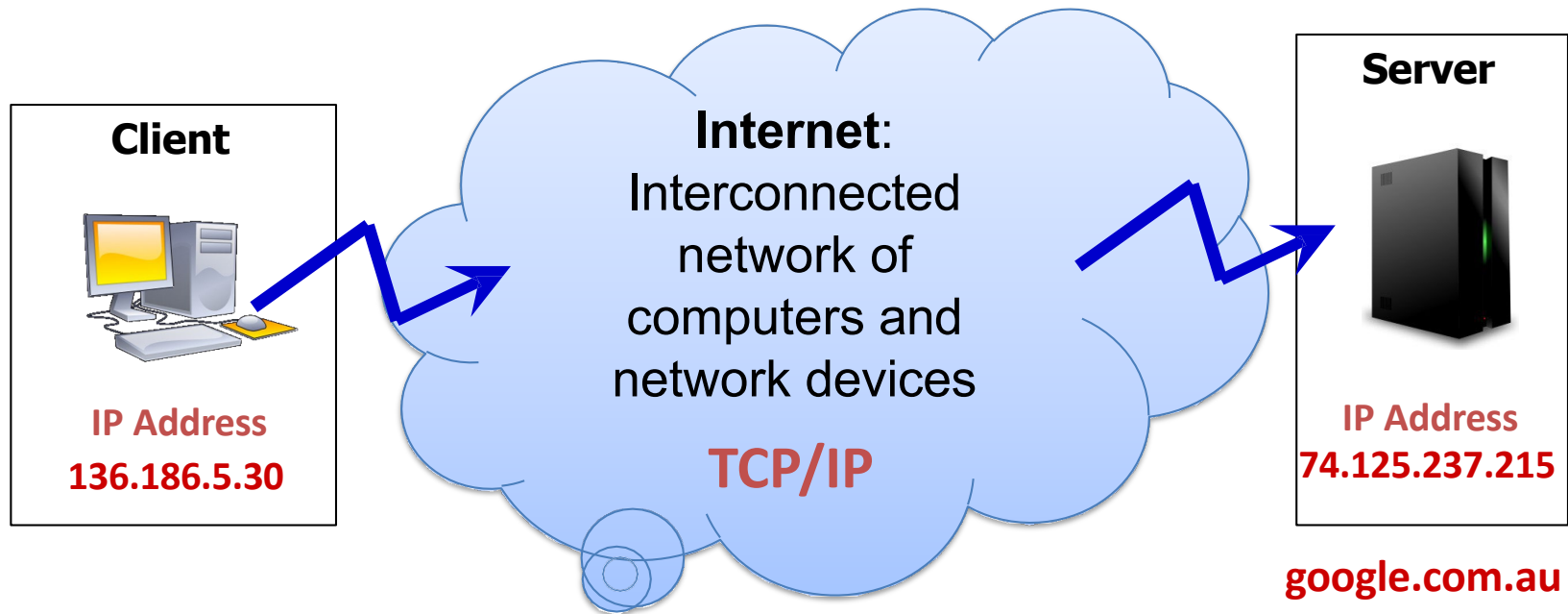
#### Introduction

# Contents

- What is Internet?
- What is the World Wide Web?
- What else is there apart from the Web?
- Understanding Web Development?
- Language versions
- Getting Started

# Internet – The Network

It is a massive network of networks



Message interception  
possible!!!

*In simple terms:*



**TCP** handles the messages  
**IP** handles the delivery

Question: Do cell phones have IP addresses?

**KNOW  
ING**

# Internet - Protocols

- To help understand how information travels over the Internet, a knowledge of the **Internet protocol suite** is needed.
- The **Internet protocol suite** is the networking model and a set of *communication protocols* used for the *Internet* and similar networks.
- It is commonly known as TCP/IP, because of its most important protocols: ***Transmission Protocol (TCP)*** and ***Internet Protocol (IP)***.
- The Internet has four abstraction layers which are used to sort all Internet protocols according to the scope of networking provided.

# Internet – Layers and Protocols

## Application

- DHCP DHCPv6 DNS FTP HTTP IMAP IRC LDAP MGCP NNTP BGP NTP POP RPC RTP RTSP RIP SIP SMTP SNMP SOCKS SSH Telnet TLS/SSL XMPP *and more*

## Transport

- TCP UDP DCCP SCTP RSVP *and more*

## Internet

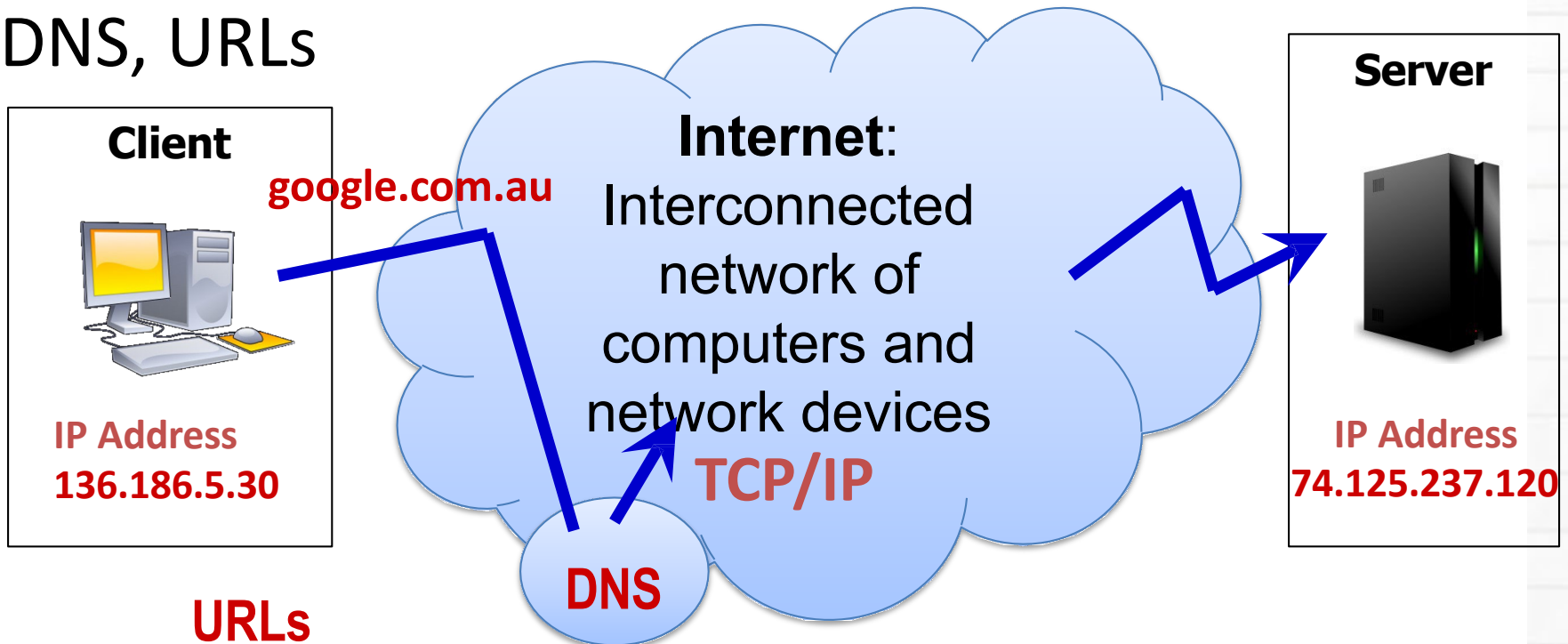
- IP (IPv4 IPv6) ICMP ICMPv6 ECN IGMP IPSE *and more*

## Link

- ARP/InARP NDP OSPF Tunnels (L2TP) PPP
- Media access control (Ethernet DSL ISDN FDDI) *and more*

# Internet – Domain Name System (DNS)

## DNS, URLs



## URLs

Internet Addresses

Look up Addresses

**google.com.au => 74.125.237.120**

**KNOW  
ING**

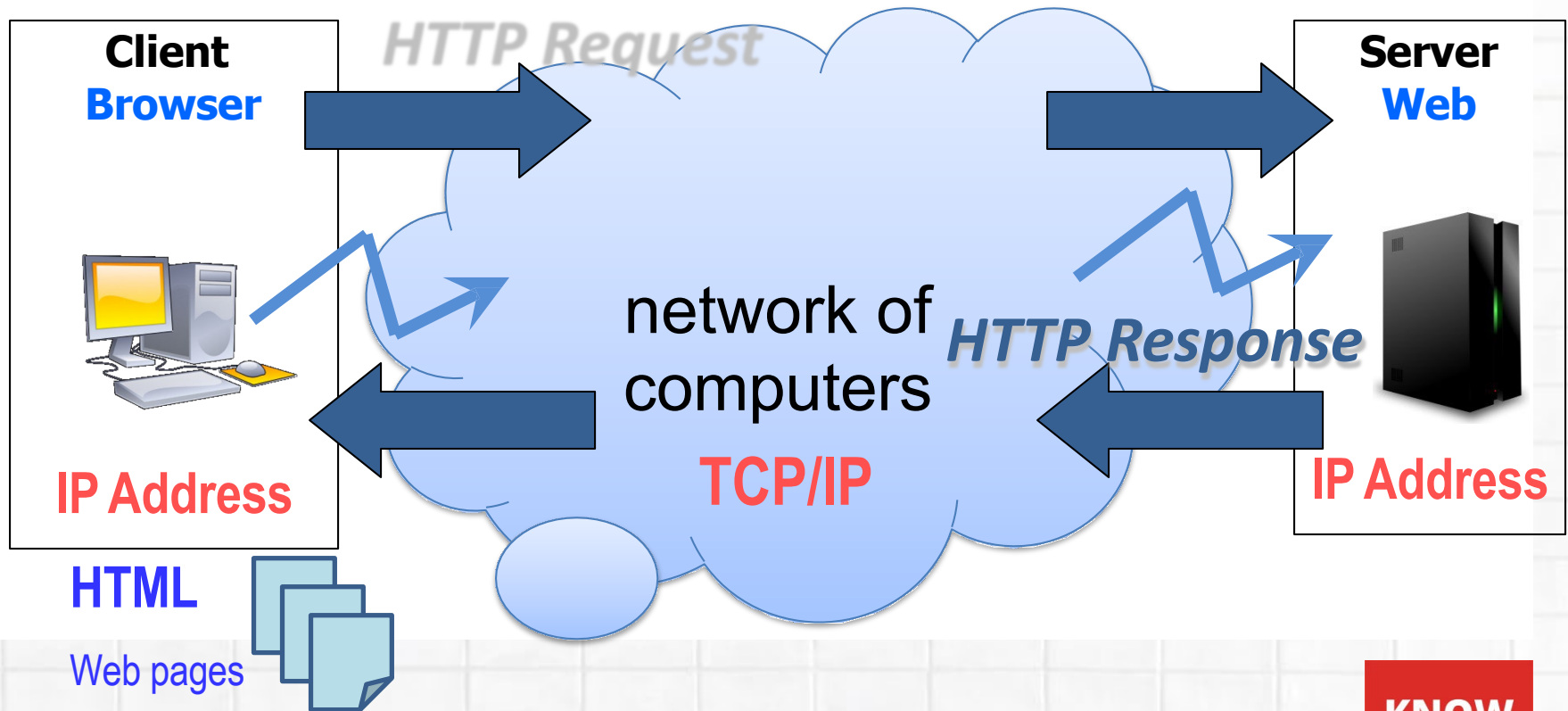
THAT WAS THE INTERNET

BUT ..

WHAT IS THE **WORLD WIDE WEB** OR THE  
**WEB?**

# The Web – The Method

A way of accessing information over the Internet.



Uses the **HTTP Protocol**

# The Web – Its Terminology

**Websites** are collections of related **Web documents** put together for a specific purpose.

The location on the Internet of the Web pages and related files

- *There are many different purposes of Web sites including scientific publication, education, ecommerce, entertainment, social networking, political campaigns etc.*

# The Web – Its Terminology

**Web Documents** – are **Hypertext** documents.

**Hypertext** is text that can act as a **link** (or hyperlink) to another document location.

- With hypertext you can skip from one topic to another in any order that suits ***you*** and ***your objectives***.
- Hyperlinks may point to another part of the same *document*, or point to another *entirely different document* (on the *same* computer or on *another* computer).
- Hyperlinks can trigger specific functions provided by the website.

# The Web – Its Terminology

**Web pages** (Web documents) are files created using:

- *HyperText Markup Language (HTML)*
  - Used to mark-up page **structure** and **content**
- *Cascading Style Sheets (CSS)*
  - Used to mark-up page **presentation**
- *JavaScript* (for client-side scripting)
  - Used to **enhance web user interaction**

# The Web – Its Terminology

**Uniform Resource Locators (URLs)** are strings used to identify the locations of unique resources on the web, e.g., web pages and image files.

- To complete the URL, we include the protocol at the start of the URL value.

<http://mercury.ict.swin.edu.au/cos10005/qhe/lab01/info.html#ex01>



The diagram shows the URL <http://mercury.ict.swin.edu.au/cos10005/qhe/lab01/info.html#ex01> with red brackets underneath it. The brackets are labeled as follows: 'protocol' for 'http://', 'server' for 'mercury.ict.swin.edu.au', 'path' for '/cos10005/qhe/lab01/info.html', 'filename' for 'info.html', and 'id' for '#ex01'.

| protocol | server                  | path                          | filename  | id    |
|----------|-------------------------|-------------------------------|-----------|-------|
| http://  | mercury.ict.swin.edu.au | /cos10005/qhe/lab01/info.html | info.html | #ex01 |

# The Web – Its Software

- **Web Browsers** (on a **personal device** as a client)
  - Send HTTP requests
  - Receive HTTP responses
  - Interpret and render/display completed Web Documents
  - *Popular Browsers: **Chrome**, **Firefox**, Safari, Opera*
- **Web Servers/HTTP Servers** (on a remote computer as a server)
  - Receive HTTP requests
  - Retrieve or create Web Documents Manage and make HTTP responses
  - *Some popular Web server software is*
    - Apache HTTP Server (Apache)***
    - Microsoft Internet Information Services (IIS) for Windows***

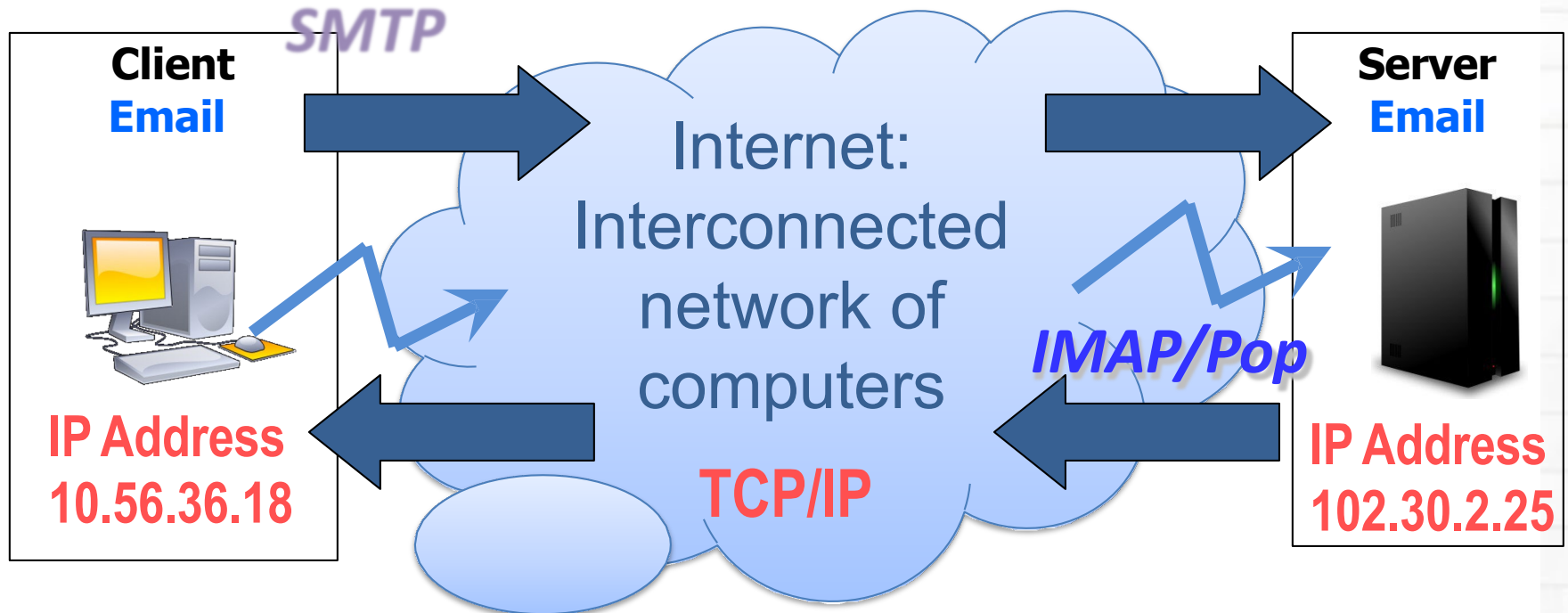
# The Web – Its Protocol

- **Hypertext Transfer Protocol (HTTP)**
  - The key “protocol” that manages the **request/response** exchange between **browsers** and **servers**
- **Hypertext Transfer Protocol Secure (HTTPS)**
  - HTTP secured with data encryption

What else is there in the Internet apart from the Web?

# What else? - Email

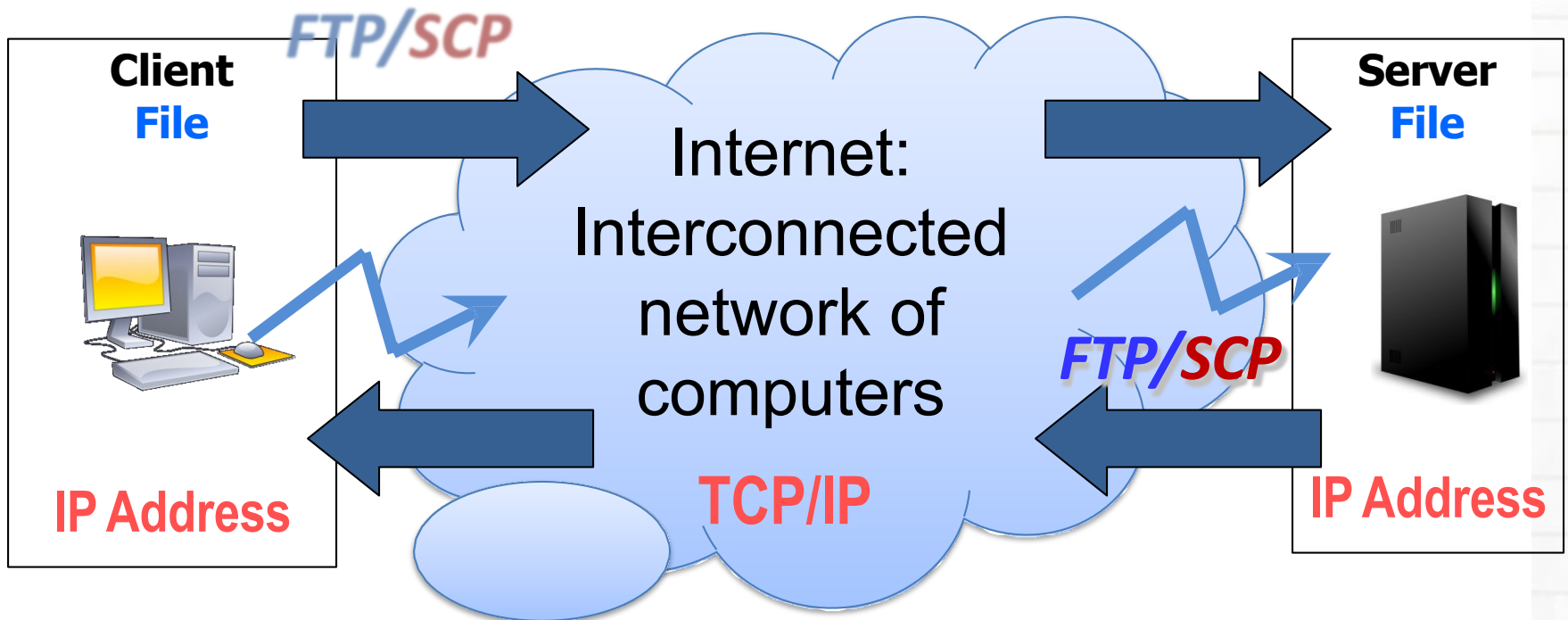
- Is a way of doing email over the Internet



- Uses the SMTP and IMAP/POP protocols.
- Popular applications: gmail.com and outlook.com

# What else? – Online Storage

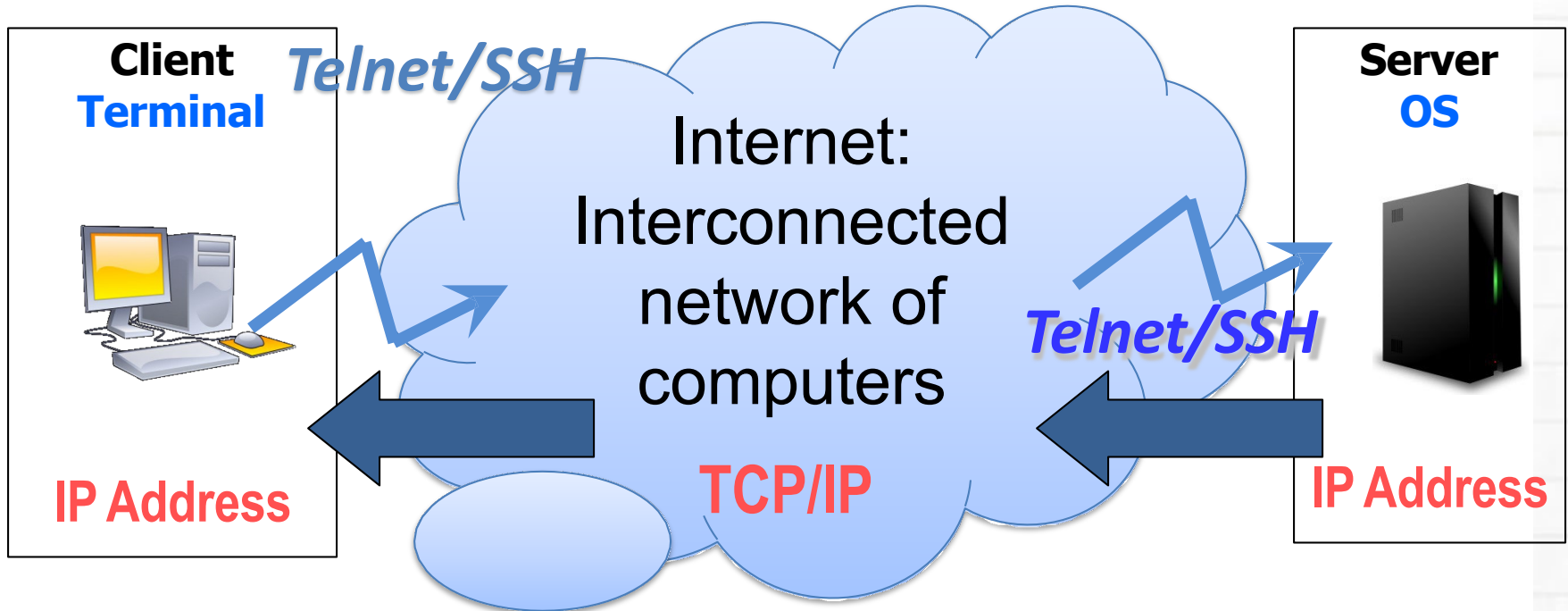
- Is a way of transferring files over the Internet.



- Protocols: FTP and SCP
- Popular applications: Dropbox, Google Drive and OneDrive

# What else? – Remote Access

- Is a way of controlling another machine over the Internet.



- Uses the Telnet or SSH protocols
- Popular applications: LogMeIn, Microsoft Remote Desktop and Chrome Remote Desktop

# Understanding Web Development

Who Should I become?

What Hardware Should I Have?

What Software Should I Install?

What Language Should I Learn?

# Web Development

- The People
- The Hardware
- The Software
- The Languages

# Web Development – The People

- **Web developers** are a programmers who specialize in the development of websites. Their focuses are:
  - Robust backbone
  - Satisfactory performance
  - Technically sound code
- **Web Designers** are people that has knowledge designing visually appealing websites. There focuses are:
  - Layout
  - Color scheme
  - User-friendliness

# Web Development

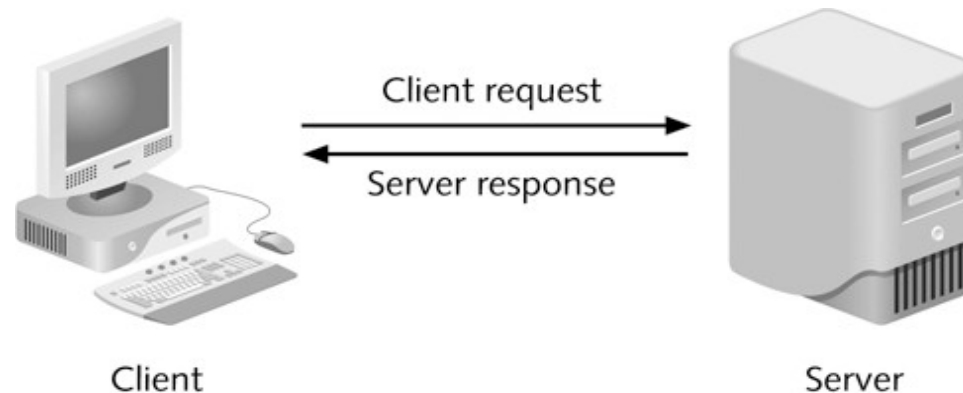
- The People
- The Hardware
- The Software
- The Languages

# Web Development – The Hardware

- **Server** ('back end')
  - Responsible for data storage and management, often has a database from which a client requests information.
  - Fulfills a request for information by managing the request or serving the requested information to the client.
- **Client** ('front end')
  - Presents an interface to the user.
  - Gathers information from the user, submits it to a server, then receives, formats, presents the results returned from the server.

# Web Development – The Hardware

- A system consisting of a client a server is known as a **two-tier** system.



**The design of a two-tier client/server system**

- Note the client and server are referring to the physical machine in this illustration.

# Web Development – The Hardware

Client tier



IP usually dynamically assigned by ISP, e.g., Telstra, Optus and TPG

Processing tier



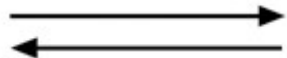
Fixed public IP  
136.186.123.37  
mercury.swin.edu.au

Data storage tier

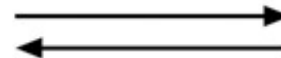


Fixed public IP  
136.186.123.38  
mysql.swin.edu.au

Client request



Server response



The design of a **three-tier** client/server system

**Safety, Consistency and Cost**

# Web Development

- The People
- The Hardware
- The Software
- The Languages

# Web Development – The Software

- **Client software** refers to the software that runs on the client machines to communicate with a server, examples are:
  - **OS** – (Windows, Linux, OSx)'s telnet, FTP
    - Third party – putty, WinSCP, CyberDuck
  - **Web Browsers** – Firefox, Chrome, Opera, Safari

# Web Development – The Software

- **Server software** refers to the software that run on the server machines, including:
  - **OS:** Windows Server, Linux Server
  - **Web:** Apache, Microsoft Internet Information Services.
  - **Database:** MS SQL Server, ORACLE, MySQL
  - **Script Support:** NodeJS (Javascript), Apache Tomcat (Java Server Pages (JSP)), Microsoft ASP (Active Server Pages), Adobe ColdFusion, Perl, Php, Python, Ruby

# Web Development

- The People
- The Hardware
- The Software
- The Languages

# Web Development – The Languages

- **HyperText Markup Language (HTML)** is a markup language designed to specify structure and content of a web page.
- **Cascading Style Sheets (CSS)** a style sheet language for describing the look and formatting (e.g., fonts, colors, spacing) of a Web documents.
- **Client-side scripting (JavaScript)** is a language often used to allow web pages to interact with users.

Content

Presentation

Interactivity

# Web Development – The Languages

- **JavaScript** is a *client-side* scripting language that is primarily used to add interactivity to web pages.
- **JavaScript** allows you to:
  - Turn static Web pages into applications, such as games or calculators. Change the contents of a Web page after a browser has rendered it.
  - Create visual effects such as animation.
  - Control the Web browser window itself.

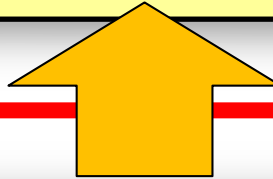
# Web Development – The Languages

## Behaviour / Interactivity

Use **scripting** to control content behaviour



Separate

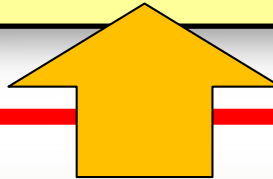


## Presentation

Use **CSS** to present the content



Separate



## Structured Content

Use **HTML / XHTML** to describe the content

# Language Versions

# Which HTML?

| Timeline  | HTML                                 | XHTML                                  |
|-----------|--------------------------------------|----------------------------------------|
| 1995      | 2.0                                  |                                        |
| 1997      | 3.2                                  |                                        |
| 1997-1998 | 4.0 (Strict, Transitional, Frameset) |                                        |
| 1999      | 4.01                                 |                                        |
| 2000      |                                      | 1.0                                    |
| 2001      |                                      | 1.1                                    |
| 2008      | 5.0 (Draft)                          |                                        |
| 2009      |                                      | 2.0 (abandoned, incompatible with 1.x) |
| 2014      | 5.0                                  |                                        |
| 2016      | 5.1                                  |                                        |
| 2017      | 5.2                                  |                                        |

# Understand HTML 4.01

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML  
4.01//EN"  
"http://www.w3.org/TR/html4/strict.dtd">  
<html>  
<head>  
<title>Title of document</title>  
</head>  
<body>  
</body>  
</html>
```

# Understand HTML5

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <meta charset="utf-8">
```

```
  <title>Title of document</title>
```

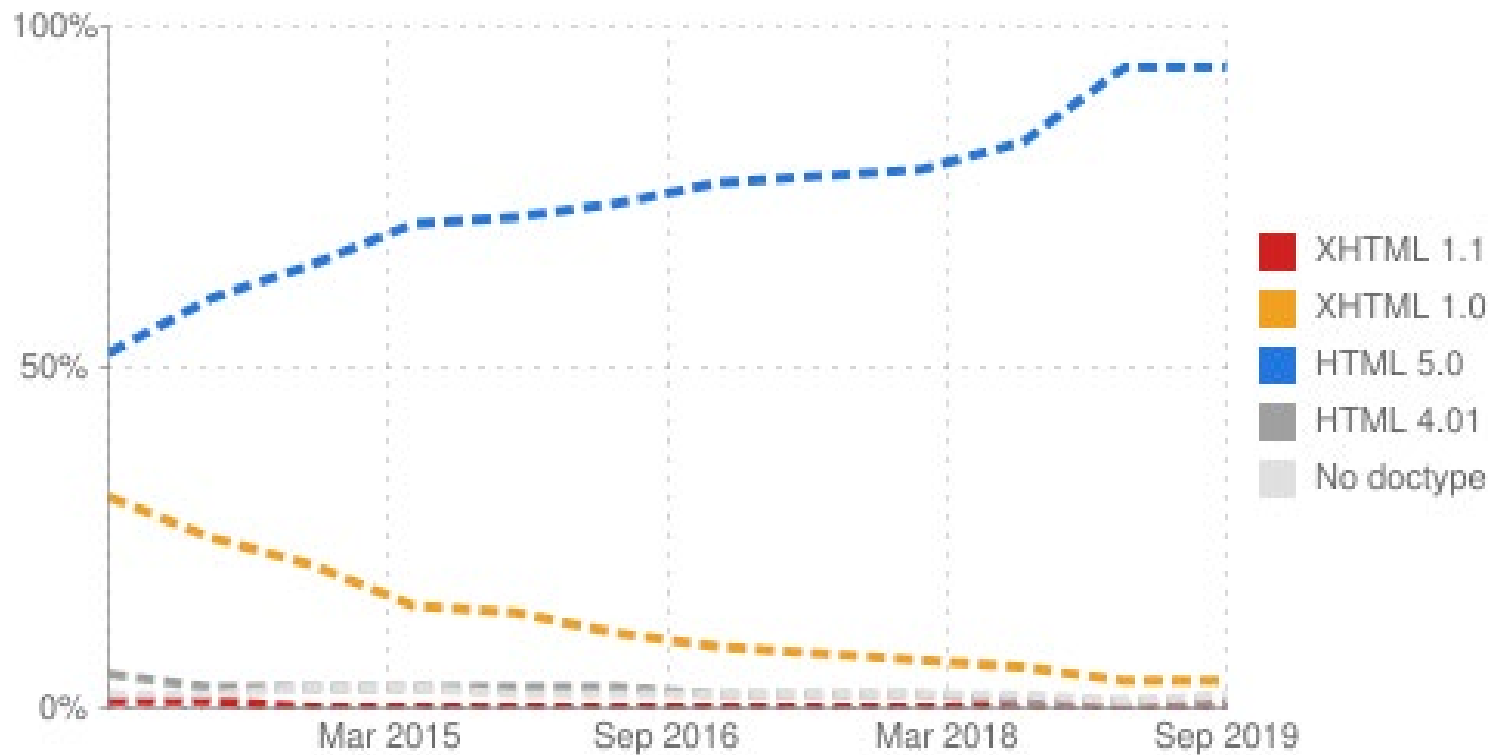
```
</head>
```

```
<body>
```

```
</body>
```

```
</html>
```

# HTML5 vs. HTML4 vs. HTML1.0



Source: <https://try.powermapper.com/stats/HtmlVersions>

# What about CSS?

| Timeline | Version | Capabilities                                         |
|----------|---------|------------------------------------------------------|
| 1996     | CSS1    | Font, Alignment, Margin, Border                      |
| 1998     | CSS2    | Positioning, z-index                                 |
| 2011     | CSS2.1  | Fixes errors in CSS2                                 |
| 2012     | CSS3    | Media Queries, Namespaces, Selectors Level 3, Colour |
|          | CSS4?   |                                                      |

# And JavaScript?

| Timeline      | Version                |
|---------------|------------------------|
| March 1996    | 1.0                    |
| August 1996   | 1.1                    |
| June 1997     | 1.2                    |
| October 1998  | 1.3                    |
|               | 1.4 (server side only) |
| November 2000 | 1.5                    |
| November 2005 | 1.6                    |
| October 2006  | 1.7                    |
| June 2008     | 1.8                    |
| February 2009 | 1.8.1                  |
| June 2009     | 1.8.2                  |
| July 2010     | 1.8.5                  |

# Getting started – Software Installation

- **Server Software**

- Apache to host your web page: XAMPP

- **Client Software**

- Web Browser: Mozilla Firefox with Web Developer add-on to view the web page.

- Text Editor: Notepad++ to edit the web page code.

# Thank You